

Windowed Quantum Phase Estimation: Signal Processing Approach to a Quantum Algorithm

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Abstract—Quantum phase estimation (QPE) is a key ingredient to some of the most important results of quantum computing. In this paper we examine the QPE algorithm through the looking glass of signal processing. This perspective helps to reveal a relation between the distribution of quantum phase estimates to the shape of the discrete Fourier transform (DFT) of a sinusoid. Guided by spectral shaping theory, we design a class of quantum circuits that are the quantum equivalents to classical windowing operators, for a family of window functions that have compact support in the DFT. These circuits have complexity linear in the number of target qubits, strictly lower than QPE itself. Using the circuits we are able to reshape QPE’s performance characteristics. In particular, the asymptotic marginal cost for 2x reduction in fail rate (2x improvement in reliability) is reduced from one qubit to 1/3 qubit or lower, which can help reduce the qubit budget for preset accuracy-reliability target, especially in high-reliability scenarios. This makes a demonstration of the value of signal processing principles in designing quantum algorithms.

Keywords—quantum algorithms, quantum phase estimation, windowed Fourier transforms

I. INTRODUCTION

Quantum computing [1] exploits quantum-mechanical effects of physical systems to build quantum circuits and algorithms that show substantial speedup against classical counterparts in suitable tasks. States of quantum circuits are processed by unitary operators. It is no surprise that a fast algorithm to the unitary eigenvalue problem [2][3], called quantum phase estimation (QPE), enjoys special importance among quantum algorithms. QPE is key to several notable quantum algorithms claiming exponential speedup over classical alternatives, including integer factoring and discrete logarithm (Shor [4]), solving linear equations (HHL [5]), fitting data by linear regression [7], quantum versions of PCA [8] and SVD [9], and quantum counting [10]. The performance characteristics of QPE has profound implications on quantum computing.

From the signal processing perspective, QPE can be understood as estimating the frequency of a pure complex sinusoid from its discrete Fourier transform (DFT). Performance of the algorithm is determined by the shape of that discrete spectrum. We are interested in looking up whether classical spectral shaping techniques can help reshape the performance characteristics of QPE. In particular, we seek a quantum algorithm that implements *windowing*, a familiar operator in spectral analysis, with exponential speedup against its classical counterpart. Since windowing sharpens spectral peak of a complex sinusoid towards the true frequency, we expect windowed QPE to yield better estimates.

This paper is arranged with signal processing audience in mind. Section 2 reviews the fundamentals of quantum computing, which we retell in subspace terminology that is already familiar in signal processing. Section 3 recounts the QPE algorithm, and explains the relation between QPE performance, spectral shape, and windowing. Section 4 presents our quantum windowing circuit, and numerically examines how windowing affects QPE in terms of tradeoffs between accuracy, reliability, and resource. We conclude with further discussions in section 5.

II. QUANTUM COMPUTING

Quantum computers hold information in quantum-mechanical devices called quantum bits, or “qubits”. Like classical bits, a qubit has two possible *eigenstates* $|0\rangle$ and $|1\rangle$. A qubit differs from a classical bit in that the qubit can be in *superposition* state, which is a normalized linear combination of eigenstates (consider eigenstates as orthonormal column vectors):

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle, a_0, a_1 \in \mathbb{C}, |a_0|^2 + |a_1|^2 = 1 \quad (1)$$

The state space of a qubit is (isomorphic to) $\mathbb{C}^2$, in which valid states are those on the unit sphere. This has two canonical subspaces which are spanned by $|0\rangle$ and $|1\rangle$, respectively. We denote these subspaces as $|j = 0\rangle$ and $|j = 1\rangle$. Here the bold j stands for the qubit. $|j = 0\rangle$ is simply the set of states in which qubit j remains in eigenstate $|0\rangle$, up to a global phase factor $e^{i\theta}$, $\theta \in \mathbb{R}$. We say that $|\psi\rangle$ has amplitude a_j , substate $a_j|j\rangle$, and energy $|a_j|^2$ in subspace $|j = j\rangle$. Total energy of a state in all subspaces must sum to 1.

Things get more interesting when there are more than one qubits. A register of m qubits has 2^m eigenstates, indexed by all 2^m combinations of individual qubits’ 0’s and 1’s. We may denote an m -qubit eigenstate either in binary form as $|b_{m-1} \dots b_1 b_0\rangle$, where $b_k \in \{0,1\}$ comes from the k^{th} qubit, or in integer form as $|j\rangle$, $j = \sum 2^k b_k$. An arbitrary state has the form $|\psi\rangle = \sum_{j=0}^{2^m-1} a_j|j\rangle$, $\sum_j |a_j|^2 = 1$. The state space is $\mathbb{C}^{2^m}$, to which the eigenstates form an orthonormal basis. Valid states are again those on the unit sphere. Notice that the dimension of the state space (hence data capacity of the register) grows exponentially with the number of qubits.

Eigenstate-aligned subspaces can be defined for an arbitrary bipartition of the m qubits. Choose a subset of n qubits, we denote this subset by j^n and its $(m - n)$ -qubit complement by $\bar{j}^n$. This split defines 2^n subspaces $|j^n = j\rangle$, $j = 0, \dots, 2^n - 1$. An orthonormal basis of subspace $|j^n = j\rangle$ consists of all 2^{m-n} eigenstates whose j^n bits evaluate to j . In each $|j^n = j\rangle$ the subset $\bar{j}^n$ remains in eigenstate $|j\rangle$, while the other qubits ($\bar{j}^n$) can be in any superposition state. The projection of a state $|\psi\rangle$ in $|j^n = j\rangle$ is its substate in this subspace, which we denote $|\psi_{j^n=j}\rangle$. To give it a concrete signal processing example, suppose we have a 1-D time signal $x = (x_0, x_1, \dots)$ of length 2^{16} encoded in the state of a 16-qubit register, i.e. $|\psi\rangle = \sum_{j=0}^{2^{16}-1} x_j|j\rangle$. Eigenstates of the 16 qubits form a complete time index from 0 to $2^{16} - 1$. Choosing the 6 highest qubits as j^n , we get $2^6 = 64$ subspaces $|j^6 = j\rangle$, $j = 0, \dots, 63$, of dimension 1024 each. This has the interpretation of splitting the whole signal into 64 disjoint time frames of length 1024, the j^{th} frame being the sequence of amplitudes of substate $|\psi_{j^6=j}\rangle$. Conversely, the 10 lower qubits $\bar{j}^6$ index another set of 1024 *dual* subspaces of dimension 64. The energy of a substate is the square sum of its amplitudes, in perfect agreement with the common notion of a signal frame’s energy.

Quantum states are processed by unitary operators called gates and circuits, which may be thought as quantum computing analogies to instructions and algorithms. A gate operating on a single qubit is just a 2×2 unitary operator $U: \mathbb{C}^2 \rightarrow \mathbb{C}^2$, which takes input $|\psi\rangle$ to

TABLE I. EXAMPLES OF SINGLE-QUBIT GATES

Name	NOT	Hadamard	R_k^*	Y-rotation
Diagram symbol				
Matrix form	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i2\pi/2^k} \end{bmatrix}$	$\begin{bmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}$

* R_k also appears in a more familiar variant as “z-rotation” gate.

output $U|\psi\rangle$. A few examples are given in Table I, in which the matrix forms are to be left-multiplied to states written as column vectors. The first rows/columns in the matrices align to eigenstate $|0\rangle$, hence the Hadamard gate takes $|0\rangle$ to $(|0\rangle + |1\rangle)/\sqrt{2}$ (“equal superposition”), $|1\rangle$ to $(|0\rangle - |1\rangle)/\sqrt{2}$, and the R_k gate leaves $|0\rangle$ unchanged but takes $|1\rangle$ to $e^{i2\pi/2^k}|1\rangle$. These gates can be chained up to form single-qubit circuits. Circuit diagrams are read from left to right, each horizontal wire representing one qubit throughout its lifespan. For example the circuit applies the R_k gate after X gate; its operator is $R_k X$, which is still unitary 2×2 .

An intuitive way to explain the power of quantum computing is by considering what happens when a single-qubit gate is applied to one qubit, say j^1 , within an m -qubit register. Intuitively, this is a unitary operator on state space $\mathbb{C}^{2^m}$ and it shall leave any eigenstate of the other $m - 1$ qubits unchanged. By the linearity of unitary operators, it is not hard to realize that the gate will remix the two subspaces $|\psi_{j^1=0}\rangle$ and $|\psi_{j^1=1}\rangle$, each of dimension 2^{m-1} , by

$$[|\psi_{j^1=0}\rangle \quad |\psi_{j^1=1}\rangle] \mapsto [|\psi_{j^1=0}\rangle \quad |\psi_{j^1=1}\rangle] U^T, \quad (2)$$

in which the subspaces are seen as column vectors, and $U \in \mathbb{C}^{2 \times 2}$ is the gate’s matrix. Equivalently, we may think the gate as being applied in all 2^{m-1} dual subspaces $|j^{m-1} = k\rangle, k = 0, \dots, 2^{m-1} - 1$, each of dimension 2. This is $O(2^m)$ parallelism by classical standards, brought about by one gate evolving an m -qubit register. In the 16-qubit example, one Hadamard gate on the highest qubit will compute the sum and difference between the first and second halves of the encoded signal. More generally, applying a circuit on a n -subset of m qubits “broadcasts” the n -qubit operator across all 2^{m-n} dual subspaces. For example, it is known that passing n qubits each through a Hadamard gate computes 2^n -point Hadamard transform in the state space. Applying this circuit with $n = 10$ on the 10 lowest qubits in our 16-qubit setup will return the Hadamard transforms of all 64 frames in substates $|j^6 = j\rangle, j = 0, \dots, 63$.

Circuits consisting of single-qubit gates only are very limited as to what states are reachable from an initial state. To make full use of the state space the qubits must *interact* via multi-qubit gates. The only important multi-qubit gate is the controlled NOT (or CNOT) gate, in the sense that all unitary circuits can be equivalently built from single-qubit gates and CNOTs at reasonable overheads. CNOT is a gate operating on two qubits, called *control* (“c”) and *target* (“t”), respectively. As shown in Fig. 1, CNOT takes eigenstate $|c, t\rangle$ to $|c, t \oplus c\rangle$, where $c, t \in \{0, 1\}$, and $\oplus$ is the binary XOR operator. The interpretation is that CNOT does nothing in subspace $|c = 0\rangle$, but applies NOT on qubit t in subspace $|c = 1\rangle$. More generally, for any circuit U one can build its controlled version C-U = $\begin{bmatrix} I & \\ & U \end{bmatrix}$ at complexity linear in U , which takes an extra qubit c for input and applies U in subspace $|c = 1\rangle$ only.

CNOT and NOT are examples of *permutation* circuits, which take each eigenstate to another eigenstate. Since unitary operators are invertible no two eigenstates are taken to the same target, so the outputs to the whole set of eigenstates must be its permutation. These circuits are said to simulate invertible classical algorithms, because they incur no new superposition. Conversely, all invertible classical algorithms can be quantumly simulated with linear overhead by a

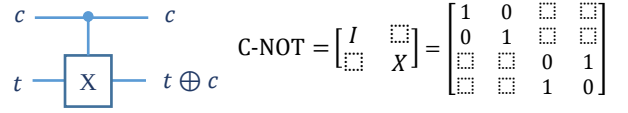


Fig. 1 Controlled NOT (CNOT) gate

permutation circuit. This allows using invertible classical subroutines in a quantum algorithm in the form of their quantum simulation. Applying a permutation circuit on a subset of qubits permutes their associated substates. Back to our 16-qubit example, a CNOT gate on the two highest qubits leaves the first half of the signal unchanged, but swaps the order of the third and fourth quarters; a classical increment operator on the top 6 qubits, i.e. $j \mapsto (j + 1) \bmod 2^6$, circularly shifts the whole signal by 1024 points.

Quantum information in qubits can be converted to classical form by quantum *measurement*. Measuring a state $|\psi\rangle$ on a subset of n qubits, say j^n , randomly reveals an eigenstate $j \in \{0, \dots, 2^n - 1\}$ as the classical outcome, with probability of seeing each j equal to the energy of substate $|\psi_{j^n=j}\rangle$. Notice this is drawing a sample from an unstructured distribution of size 2^n in time $O(n)$. Depending on the outcome j , the system’s state after measurement becomes its projection in subspace $|j^n = j\rangle$, i.e. $|\psi_{j^n=j}\rangle$, renormalized to unit energy. In our 16-qubit example, measuring the 6 higher qubits j^6 randomly chooses a frame (prioritizing the louder ones), puts all energy in that frame, and makes all other frames zero. Besides retrieving the outcome, measurement is also used for its subspace projection capability to build non-unitary operators, as with the HHL algorithm [5]. In both cases the randomness involved may incur some chance of failure. Detectable failures result in wasted compute, which reduces an algorithm’s *efficiency*; undetected failures lead to wrong results, which corrupts the algorithm’s *reliability*. Measurement is drawn in circuit diagrams by the symbol , using a double wire to mark the output as a classical bit.

In summary, quantum computing offers data capacity, parallelism, and sampling speed that are exponential in the number of qubits. These are the main sources of quantum advantage against classical computing. Consequently, one need use sufficiently many qubits to achieve quantum advantage.

III. QUANTUM PHASE ESTIMATION AND ITS SIGNAL PROCESSING INTERPRETATION

Given unitary operator U and its eigenvector $|u\rangle$, QPE computes the corresponding eigenvalue $\lambda = e^{i\phi}$, or equivalently ϕ (“phase”), in classical form. The classical difficulty of the problem lies in the $O(2^m)$ size of the state space. The QPE algorithm proceeds in three steps: 1) use U and $|u\rangle$ to prepare a pure complex sinusoid $x_j = e^{i\phi j}, j = 0, \dots, 2^m - 1$, in an m -qubit target register; 2) quantumly compute the discrete Fourier transform (DFT) of this sinusoid in place; 3) measure the register to obtain outcome $\hat{k}$, and output estimate $\hat{\phi} = 2\pi\hat{k}/2^m$. Of the three steps, step 2 is well understood to be $O(m^2)$, thanks to a quantum Fourier transform circuit that does the job using m Hadamard and $m(m - 1)/2$ controlled R_k gates, while step 3 is $O(m)$. Complexity of step 1 largely depends on the operator U . Several “best case” use cases [4][5] have bounded step 1 to $O(m^2)$, so that the overall complexity is comparable to quantum DFT. We are interested in applying windowing to the Fourier transform step without significant overhead. It needs be at most $O(m^2)$ to be applicable to general QPE use cases.

This windowing is motivated by observing steps 2, 3 of the QPE algorithm as estimating the frequency of sinusoid $x_j = e^{i\phi j}$ from its DFT, seeing j as “time”. The DFT is computed in step 2 as the state of the m -qubit target register; the energy in the k^{th} DFT bin being that of eigenstate $|k\rangle$. By the rules of quantum measurement, step 3 draws a bin from the energy distribution over all bins, which is the

energy spectrum. Since we know from classical signal analysis that windowing concentrates spectral energy towards true frequency, it is natural to expect tighter estimates from windowed QPE.

More specifically, let $|x\rangle \in \mathbb{C}^{2^m}: x_j = e^{i\phi j}, \phi = 2\pi f$ be the sinusoid in the target register. We choose a window function $|w\rangle \in \mathbb{C}^{2^m}$, and feed their element-wise product $|wx\rangle$ to the quantum DFT. The energy spectrum from this DFT is given by

$$|\hat{x}_k^w|^2 = |\hat{w}(k/M - f)|^2, k = 0, \dots, M-1, \quad (3)$$

where $M = 2^m$, $\hat{w}(f), f \in \mathbb{R}$ is the discrete-time Fourier transform (DTFT) of $|w\rangle$, defined by $\hat{w}(f) := \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} w_j e^{-i2\pi j f}$. Windowed QPE draws $\hat{k}$ from the distribution $P(k) = |\hat{x}_k^w|^2$. Typical window functions in spectral analysis have their energy concentrated about a main peak at some $\delta_{pk}^w \in (-1/2, 1/2)$ bins; real symmetric windows have $\delta_{pk}^w = 0$. The quantum phase estimate using a window of arbitrary δ_{pk}^w is given by $\hat{\phi} = 2\pi(\hat{k} - \delta_{pk}^w)/M$.

Fixing m , the performance of QPE is characterized by a tradeoff between accuracy and reliability. We say QPE meets performance target (n, ϵ) if it attains n -bit accuracy with probability at least $1 - \epsilon$ for all $f \in [0, 1)$. Given f , the probability of failing to attain n -bit accuracy is explicitly

$$\epsilon_n(f) = \sum_{k=K_1^{(n)}}^{K_2^{(n)}} |\hat{w}(k/M - f)|^2, \quad (4)$$

$K_1^{(n)} = \lceil Mf + \delta_{pk}^w + 2^{m-n-1} \rceil$, $K_2^{(n)} = K_1^{(n)} + 2^m(1 - 2^{-n}) - 1$. Equation (4) sums over frequencies uniformly placed in the tails of $|\hat{w}|^2$ cut off at $\pm 2^{m-n-1}$ bins from the main peak. Standard QPE uses a rectangular window, which has notoriously long tails in the DTFT, and consequently high fail rates. It is possible to cut the fail rate by half, asymptotically, by adding one more qubit to the target register. We propose to reduce the fail rate by using windows with shorter tails. The goal is to do so at complexity no more than $O(m^2)$. We notice that the previous attempt at quantumly computing windowed Fourier transforms [12] did not meet this goal, as it had relied on naively applying a generic state preparation routine [13] at complexity at least $O(2^m)$.

IV. QUANTUM WINDOWING CIRCUIT AND ITS USE IN QPE

Fast quantum windowing can be achieved with a special class of window functions, the cosine family, whose DFTs have compact support about zero frequency. These have the explicit form

$$w_j := \frac{1}{z_c \sqrt{M}} \sum_{r=R_-}^{R_+} c_r e^{i2\pi r j / M}, j = 0, \dots, M-1. \quad (5)$$

$R_- \leq 0, R_+ \geq 0$ are small integers, $c_r \in \mathbb{C}$ is the r^{th} bin of the DFT, and $z_c = \sqrt{\sum_r |c_r|^2}$ a normalizer. These windows have ‘lobes’ which are narrow intervals between regularly appearing zeros in the DTFT, with the main lobe striding $R_- - 1$ to $R_+ + 1$ bins. Hann, Hamming and Blackman windows are examples from this family. By polynomial factorization we can rewrite the above as

$$w_j = \frac{1}{z_d} \prod_{r=R_-}^{-1} (1 - d_r e^{-i2\pi j / M}) \prod_{r=1}^{R_+} (1 - d_r e^{i2\pi j / M}) \quad (6)$$

Fig. 2 gives the quantum circuit for computing DFT with this family of windows. The circuit uses $R := R_+ - R_-$ auxiliary qubits $a_{R_-}, \dots, a_{-1}, a_1, \dots, a_{R_+}$, initialized to $|0\rangle$. The single-qubit $U(d)$ gates are parameterized by the values in (6). $FT^\dagger$ is the quantum DFT circuit. The $R_m^{\pm j}$ gates are linear phase shifters, mapping amplitudes of $|x\rangle$ by $x_j \mapsto x_j e^{\pm i2\pi j / M}$. Fig. 3 shows how the controlled version of R_m^j is built from m controlled R_k gates. We explain the circuit in Fig. 2 by showing that the subcircuit in dashed frame (bottom right) computes DFT of $|x\rangle$ with a window $w_j \propto 1 - d_{R_+} e^{i2\pi j / M}$, the last factor in (6); the rest is just nested application of the same construct. Starting with input $|x\rangle$ in the target register, the addition of auxiliary

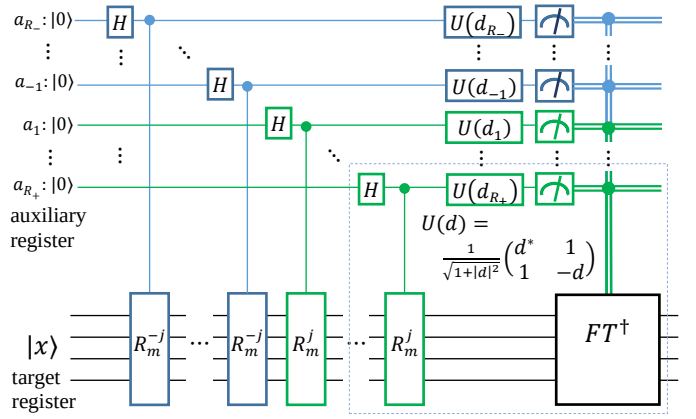


Fig. 2 Quantum circuit of windowed DFT

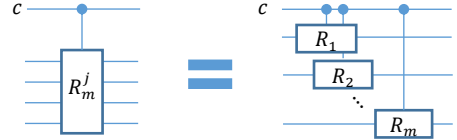


Fig. 3 Controlled linear phase shifter

qubit a_{R_+} in state $|0\rangle$ doubles the dimensionality of the state space, putting $|x\rangle$ in $|a_{R_+} = 0\rangle$, nothing in $|a_{R_+} = 1\rangle$. The Hadamard gate on a_{R_+} takes it to equal superposition of $|0\rangle$ and $|1\rangle$, effectually leaving identical copies of $|x\rangle$ in both subspaces. Controlled by a_{R_+} , the linear phase shifter maps $x_j \mapsto x_j e^{i2\pi j / M}$ in subspace $|a_{R_+} = 1\rangle$, while leaving $|x\rangle$ unchanged in $|a_{R_+} = 0\rangle$. The gate $U(d_{R_+})$ on a_{R_+} remixes the two substates, leaving correctly windowed signal $x_j - d_{R_+} x_j e^{i2\pi j / M}$ in $|a_{R_+} = 1\rangle$. Conditioned on measuring a_{R_+} to 1, this windowed substate renormalizes to become the full state in the target register. The quantum DFT then completes the job.

The windowing circuit (without DFT) uses R Hadamard gates, R $U(d)$ gates, and mR controlled R_k gates. This is strictly lower complexity than quantum DFT with $m(m-1)/2$ controlled R_k gates, for any m large enough to show quantum advantage. We note that for real-valued d the $U(d)$ gate reduces to rotation; for $d = 1$ it reduces to the Hadamard gate. Efficiency-wise the windowed DFT meets detectable failure if any of the R auxiliary qubits measure to 0. The conditioning on measurements is needed because windowing operators are generally non-unitary. We estimate the probability of measuring all auxiliary qubits to 1 to be between 0.16 and 0.5 for a typical set of windows ($P(1)$ column in Table II).

A. Numerical results

We study the effect of windowing on QPE by evaluating the performance profile expressed by (4) for several common cosine family windows listed in Table II. Direct evaluation of (4) becomes unwieldy for large m 's due to exponential data size, so we seek to approximate (4) by numerical integration. To do so we factor out the faster oscillations from $|\hat{w}(k/M - f)|^2$:

$$|\hat{w}(k/M - f)|^2 = \frac{\sin^2 \pi M f}{z_c^2 M^2} \cdot |W(k)|^2, \quad (7)$$

where $W(k) := \sum_r c_r e^{-i\pi \frac{r}{M}} \csc \pi \left(\frac{k-r}{M} - f \right)$.

Equation (7) is valid outside the main lobe. With $W(k)$ a smooth function and $\sin^2 \pi M f / z_c^2 M^2$ independent of k , we are able to approximate (4) by integrating $|W|^2$ over $(K_1^{(n)} - 0.5, K_2^{(n)} + 0.5)$ using an off-the-shelf integrator. Calculations are carried out at 240-bit precision, which is sufficient for computing $\epsilon_n(f)$ with window types listed in Table II and m up to 64. We let $\epsilon_n = \sup_f \epsilon_n(f)$.

Table I Setup, attributes, and QPE results of evaluated window types

Window type	$c^{\diamond\diamond}$ (c_0 underscored)	d ($d_{r<0}; d_{r>0}$)	Side lobe energy (%)	Spectral rolloff	δ_{pk} (bins)	$R^{\diamond}$	$P(\mathbf{1})^{\diamond}$	ϵ_{m-1} (%)	ϵ_{m-5} (%)	ϵ_n 's rolloff $^{\diamond\diamond\diamond}$
Rectangular	<u>1</u>	- ; -	9.7	6dB/oct $\square$	0	0	-	0.19	0.013	2x/bit
Half sine	<u>1</u> , -1	- ; 1	0.5	12dB/oct	0.5	1	0.5	0.099	8.3E-6	8x/bit
Hann	-1, <u>2</u> , -1	1; 1	0.05	18dB/oct	0	2	0.375	0.17	2.6E-8	32x/bit
Hamming	-23, <u>54</u> , -23	0.559; 0.559	0.04	6dB/oct	0	2	0.341	0.13	2.0E-4	2x/bit
Hann ^{1.5}	-1, <u>3</u> , -3, 1	1; 1, 1	0.007	24dB/oct	0.5	3	0.313	0.22	2.0E-10	128x/bit
Hann ²	1, -4, <u>6</u> , -4, 1	1, 1; 1, 1	0.001	30dB/oct	0	4	0.273	0.26	2.7E-12	512x/bit
Blackman	4, -25, <u>42</u> , -25, 4	0.25, 1; 1, 0.25	0.0002	18dB/oct	0	4	0.165	0.22	4.0E-9	32x/bit

$^{\diamond} R$: number of auxiliary qubits, equals the main lobe width minus 2. $P(\mathbf{1})$: probability of measuring auxiliary bits to all 1's.

$^{\diamond\diamond}$ Scaling all elements of c by a positive number results in the same window.

$^{\diamond\diamond\diamond}$ Asymptotic scaling of fail rate for each additional bit of accuracy (roughly holds up to $m - 5$).

$\square$ "oct" stands for octave. Spectral roll-off of $3 \times p$ db/oct corresponds to $1/f^p$ -like decay of energy spectrum.

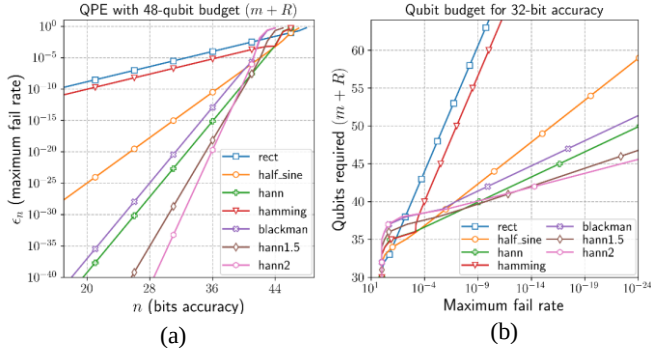


Fig. 4 Numerical simulation of QPE: (a) performance characteristics at 48-qubit budget; (b) qubit budget for 32-bit accuracy at various fail rate bounds

The main results are shown in Fig. 4. Fig. 4(a) compares QPE performance profile (n, ϵ) at 48-qubit budget (including target and auxiliary qubits) across window types; Fig. 4 (b) compares resource-reliability curves, i.e. number of qubits needed to achieve specified fail rate, fixing 32-bit accuracy. The two plots closely mirror each other, which is the result of fixed 1:1 “exchange rate” between target qubits and bit precision of estimation. The curves consist of a head part where different windows show variable behaviours, and a tail part where all curves are asymptotically log-linear. The asymptotic slopes can be analytically derived from Fourier analysis theory, which we list in the last column of Table II. We also list the upper bounds of fail rates at $(m-1)$ and $(m-5)$ bits accuracy. Empirically we observe that the log-linear rule gives good approximation of ϵ_n for $n \leq m-5$, which can be useful for quick estimation of the performance profile, e.g. during dynamic planning of algorithms and resources.

An immediate observation from the results is that windowing changes QPE’s performance characteristics. In terms of resource-reliability tradeoff, the asymptotic marginal cost for achieving 2x reliability (i.e. half fail rate), as visualized by the slopes of curves in Fig. 4 (b), is reduced from one qubit to 1/3, 1/5, 1/7 or 1/9 qubit, depending on the window type. Windowing does not affect the asymptotic cost of additional accuracy, which remains one qubit per bit accuracy. Fig. 4 also shows when windowing helps reduce overall qubit budget of QPE. Roughly speaking, rectangular (no windowing) has the lowest qubit budget for fail rates above 0.1; the half sine window has it for fail rates between 0.1 and 0.0003; then the Hann window takes over for fail rates down to 2×10^{-8} . Practical window selection will also consider factors like complexity and efficiency, but the big picture is clear: use a window if highly reliable estimation is desired.

V. CONCLUSION AND DISCUSSIONS

In this paper we have identified a link between the quantum phase estimation algorithm and classical frequency estimation from DFT, which immediately suggests using spectral shaping in QPE.

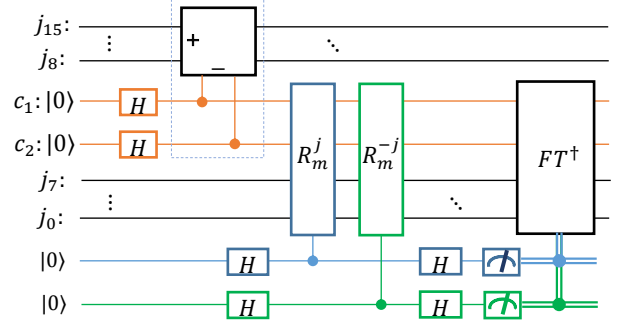


Fig. 5 Quantum STFT with Hann window and 75% frame overlap

This is accomplished by means of a quantum circuit that implements windowing in the quantum state space. By theoretical calculations we have confirmed that the method allows control of QPE’s performance profile in terms of accuracy-reliability-resource tradeoffs, and can help reduce qubit budget in high-reliability use cases. We have also included a brief review of quantum computing basics in subspace terminology to help set up context for our discussion. For a more comprehensive introduction to the field in full quantum information vocabulary we suggest [1] for a good start.

Quantum computing and signal processing have largely gone separately ways in the past decades. Only recently did their paths cross, often in the form of using quantum circuits as alternative signal transformation and optimization resources [14][15][16], or that of applying signal processing methods in classical form to address various needs of quantum computing tasks [17][18][19], including alternative methods for the QPE task [20]. What we present in this paper is a demonstration of the interdisciplinarity working another way, in which signal processing theory directly motivates a new quantum circuit design. We expect more signal-inspired quantum algorithms to come up as quantum computing gains wider attention from signal processing communities.

Since windowing is a classic signal analysis tool by itself, an interesting question is whether the windowing circuit presented here will be useful for quantum signal processing tasks, such as quantum time-frequency analysis. For now the short answer is that we have yet to confirm a use case in which doing so brings concrete end-to-end advantage compared to classical computing. Were such a use case to arise someday, doing the windowing can be straightforward. For example, the circuit in Fig. 5 computes the short-time Fourier transform (STFT) of a 2^{16} -long input encoded in qubits $j_{15} \dots j_0$, using 1024-point Hann windows with 75% overlap. The submodule in the dashed frame is a permutation circuit that simulates classical invertible subtraction $|j, c\rangle \mapsto |j - c, c\rangle$, where the deduction is modulo 2^8 . There will be time alias between frames at the front and back because of the modular arithmetic, which can be eliminated by leaving the last 75% frame empty.

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Supplementary material

The supplementary material includes proofs and derivations related to but not included in the main body of the paper “Windowed quantum phase estimation: signal processing approach to a quantum algorithm.”

A. Windowing circuit in quantum computing terminology

The following circuit implements windowing operator $x_j \mapsto x_j^w \propto x_j w_j^c$, with

$$w_j = \frac{1}{z_d} \prod_{r=R_-}^{-1} (1 - d_r e^{-i2\pi j/M}) \prod_{r=1}^{R_+} (1 - d_r e^{i2\pi j/M}) \quad (1)$$

where z_d is a normalizer independent of j .

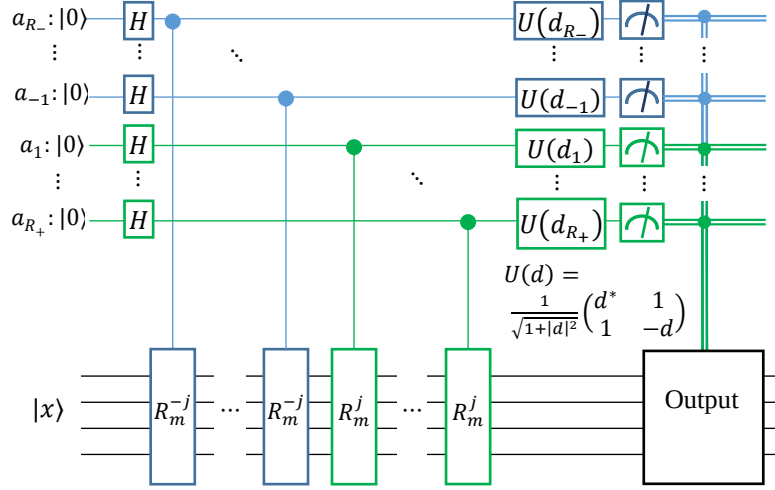


Figure 1 The windowing circuit

(Anatomy of the circuit) Let $R = R_+ - R_-$. The circuit consists of $R + 1$ parts, including a base that contains the target register $|x\rangle$ and R subcircuits associated with auxiliary qubits $a_{R_1}, \dots, a_{-1}, a_1, \dots, a_{R_+}$ respectively. Each of the R subcircuits consists of its auxiliary qubit, a Hadamard gate, a $U(d)$ gate, and measurement on that qubit, alongside a linear phase shifter $R_m^{\pm j}$ controlled by the qubit. The output port represents whatever downstream algorithm that receives the windowed input from the target register. For example, it might be the quantum DFT circuit in a windowed DFT task.

The R subcircuits work independent of each other. We only show that the subcircuit associated with qubit $a_{\pm r}$ ($r > 0$) implements windowing $x_j \mapsto x_j w_j^r$ with $w^r \propto 1 - d_{\pm r} e^{\pm i2\pi j/M}$; the full proof follows by induction.

WLOG, we consider the subcircuit with qubit a_1 . Its circuit diagram (including the base) is:

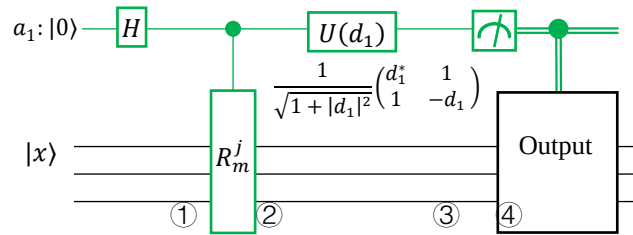


Figure 2 Subcircuit of Figure 1 with one auxiliary qubit

The initial state is $|0, x\rangle$, in which 0 is from qubit a_1 , $|x\rangle = \sum_j x_j |j\rangle$ is the initial state of the target register. The Hadamard gate takes a_1 to equal superposition, so the state becomes ①: $\frac{1}{\sqrt{2}} |0, x\rangle + \frac{1}{\sqrt{2}} |1, x\rangle$. The a_1 -controlled R_m^j gate applies linear phase to $|x\rangle$ in conditioned on $a_1 = 1$, reaching ②: $\frac{1}{\sqrt{2}} |0, x\rangle + \frac{1}{\sqrt{2}} |1, x e^{i2\pi j/M}\rangle$, in which we have used the shorthand $|x e^{i2\pi j/M}\rangle := \sum_j x_j e^{i2\pi j/M} |j\rangle$. Gate $U(d_1)$ remixes the two parts by taking $|0, x\rangle$ to $\frac{1}{\sqrt{1+|d_1|^2}} (d_1^* |0, x\rangle + |1, x\rangle)$, $|1, x\rangle$ to $\frac{1}{\sqrt{1+|d_1|^2}} (|0, x\rangle - d_1 |1, x\rangle)$, so the full state is taken to ③: $\frac{1}{\sqrt{2}} \frac{1}{\sqrt{1+|d_1|^2}} (d_1^* |0, x\rangle + |1, x\rangle + |0, x e^{i2\pi j/M}\rangle - d_1 |1, x e^{i2\pi j/M}\rangle) = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{1+|d_1|^2}} |0, x(d_1^* + e^{i2\pi j/M})\rangle + |1, x(1 - d_1 e^{i2\pi j/M})\rangle$. Conditioned on measuring $a_1 = 1$, the output port receives ④ $\propto |x(1 - d_1 e^{i2\pi j/M})\rangle$, i.e. $|x\rangle$ windowed by $w^r \propto 1 - d_1 e^{i2\pi j/M}$. $\square$

B. Controlled linear phase shifter circuit

The following circuit implements controlled linear phase shifter $x_j \mapsto x_j e^{i2\pi j/2^m}$.

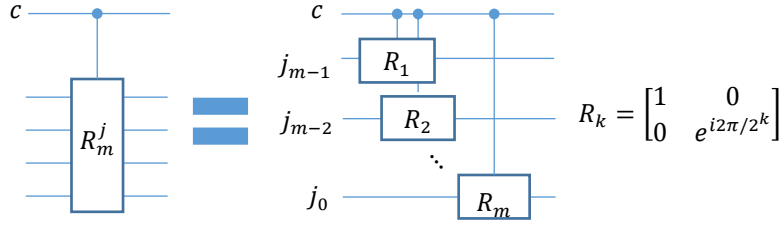


Figure 3 Controlled linear phase shifter

Fix a basis vector $|j\rangle$ of the target register. Let $|j_{m-1} \dots j_1 j_0\rangle$ be its binary encoding, i.e. $j = \sum_{k=0}^{m-1} 2^k j_k$. For each $k = 0, \dots, m-1$, the qubit j_k goes through gate $R_{m-k} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i2\pi \cdot 2^k/2^m} \end{pmatrix}$. This applies a phase $2\pi 2^k/2^m$ if $j_k = 1$, and has no effect if $j_k = 0$. In either case this gate incurs a phase shift $(2\pi/2^m)2^k j_k$ to $|j\rangle$. The total phase shift through all the m R_{m-k} gates is their sum $\sum_{k=0}^{m-1} \frac{2\pi}{2^m} 2^k j_k = \frac{2\pi}{2^m} \sum_{k=0}^{m-1} 2^k j_k = \frac{2\pi}{2^m} j$, hence the map $x_j \mapsto x_j e^{i2\pi j/2^m}$. $\square$

C. Unitary operator applied on a subset of qubits

Consider applying an n -qubit unitary operator (i.e. quantum circuit) $U: \mathbb{C}^{2^n} \rightarrow \mathbb{C}^{2^n}$ on an n -subset, $\mathbf{j}^n$, of a m -qubit register. We can arrange the 2^m elements of the state vector in a $2^n \times 2^{m-n}$ matrix $\Psi \in \mathbb{C}^{2^n \times 2^{m-n}}$, the j^{th} row containing amplitudes of eigenstates whose $\mathbf{j}^n$ subset of the index evaluate to j , the k^{th} column containing amplitude of eigenstates whose $\mathbf{j}^n$ subset of the index evaluates to k . Rows of the matrix form the subspaces indexed by $\mathbf{j}^n$; columns of the matrix form the dual subspaces indexed by $\mathbf{j}^n$. Fixing a state vector $|\psi\rangle$ in the register, the j^{th} row of the matrix contains substate $|\psi_{j^n=j}\rangle$.

Now U acting on the m -qubit register becomes a linear operator on $\mathbb{C}^{2^n \times 2^{m-n}}$ independent of the columns. This must have the form

$$U: \Psi \mapsto \tilde{U}\Psi. \quad (2)$$

That $U = \tilde{U}$ can be established by considering states in $|\mathbf{j}^n = 0\rangle$, i.e. $|\psi\rangle = |\psi_{j^n=0}\rangle$, in which $|\psi_{j^n}\rangle$ is both the first column of Ψ and a valid n -qubit state by itself. $\square$

D. DTFT of window functions in the cosine family

Let the window function be $w_j = \frac{1}{z_c \sqrt{M}} \sum_{r=R_-}^{R_+} c_r e^{i2\pi r j/M}$. By definition its DTFT is $\hat{w}(f) = \frac{1}{\sqrt{M}} \sum_j w_j e^{-i2\pi j f} = \frac{1}{z_c M} \sum_r c_r \sum_j e^{-i2\pi j(f-r/M)}$. We have $\sum_{j=0}^{M-1} e^{-i2\pi j f} = M e^{-i\pi(M-1)f} \text{sinc}(Mf; M)$, where $\text{sinc}(x; M) := \lim_{x' \rightarrow x} \frac{\sin \pi x'}{M \sin(\pi x'/M)}$ is the discrete sinc function of order M . Then

$$\hat{w}(f) = \frac{1}{z_c} \sum_r c_r e^{-i\pi(M-1)(f-r/M)} \text{sinc}(M(f-r/M); M) = \frac{1}{z_c} e^{-i\pi(M-1)f} \sum_r c_r (-1)^r e^{-i\pi r/M} \text{sinc}(Mf-r; M) \quad (3)$$

Taking the limit in the definition of $\text{sinc}(x; M)$ is needed to address zero denominators, i.e. the case $x = kM, k \in \mathbb{Z}$. If f stays clear of the main lobe, we can safely write $\text{sinc}(Mf-r; M) = \frac{\sin \pi(Mf-r)}{M \sin \pi(f-r/M)} = \frac{(-1)^r}{M} \frac{\sin \pi Mf}{\sin \pi(f-r/M)}$, and

$$\hat{w}(f) = \frac{\sin \pi Mf}{z_c M} e^{-i\pi(M-1)f} \sum_r c_r e^{-i\pi r/M} \csc \pi(f-r/M) \quad (4)$$

where $\csc x := 1/\sin x$. Notice this expression has factored out the fast oscillating term $\sin \pi Mf$ clearly of the DFT bin index, i.e. r . $\square$

E. QPE's fail rate, i.e. the probability of not meeting some accuracy requirement

QPE produces an estimate by randomly sampling an integer $\hat{k}$ from distribution $|\hat{x}_k^w|^2 = |\hat{w}(k/M - f)|^2$. The quantum phase is estimated as $\hat{\phi} = 2\pi(\hat{k} - \delta_{pk}^w)/M$, or, the frequency is estimated as $\hat{f} = (\hat{k} - \delta_{pk}^w)/M$. Here δ_{pk}^w is the position of the window function's DTFT peak measured in DFT bins. $\delta_{pk}^w = 0$ for rectangular, Hann, Hamming, Hann² and Blackman windows; $\delta_{pk}^w = 0.5$ for half sine and Hann^{1.5} windows.

We say a measurement of f meets n -bit accuracy if $\hat{f} - f \in \left[-\frac{1}{2^{n+1}}, \frac{1}{2^{n+1}}\right)$, or $\hat{k} - \delta_{pk}^w - Mf \in \left[-\frac{2^m}{2^{n+1}}, \frac{2^m}{2^{n+1}}\right)$. Taking the periodicity of $\hat{x}_k^w = \hat{x}_{k+M}^w$ into account, the condition of failing n -bit accuracy can be expressed as $\hat{k} - \delta_{pk}^w - Mf \in \left[\frac{2^m}{2^{n+1}}, 2^m - \frac{2^m}{2^{n+1}}\right)$. The first and last integers in this range are

$$K_1 = \lceil Mf + \delta_{pk}^w + 2^{m-n-1} \rceil, K_2 = \lceil Mf + \delta_{pk}^w + 2^m - 2^{m-n-1} \rceil - 1. \quad (5)$$

The probability of failing n -bit accuracy is then $\epsilon_n(f) = \sum_{k=K_1}^{K_2} |\hat{x}_k^w|^2$. For window functions of the cosine family, assuming that range $[K_1, K_2]$ does not intersect with the main lobe, we have the approximation

$$\epsilon_n(f) \cong \frac{\sin^2 \pi M f}{z_c^2 M^2} \int_{K_1-0.5}^{K_2-0.5} \left| \sum_r c_r \frac{e^{-i\pi r/M}}{\sin \pi((k-r)/M-f)} \right|^2 dk \quad (6)$$

F. Asymptotic spectral decay of cosine family windows

Table 2 below lists the discrete-time spectra of the rectangular, half sine and sine windows in closed form, assuming non-zero denominators, which holds outside the main lobe. Each spectrum is explicitly factored into a linear phase, a normalizer, a smooth shape factor, and a fast oscillation $\sin \pi M f$. The laws of spectral decay follow directly from the number of sine factors in the denominator of the shape factor. Since most spectral energy is concentrated in low frequencies, where the sine function is approximately linear, the decay laws are approximately polynomial in the range we care about. Notice that the decay laws given are of the spectral magnitude. $1/f^n$ decay in magnitude translates to $1/f^{2n}$ decay in energy density, and to $1/f^{2n-1}$ decay in total energy in the spectral tail cut off at f . This explains the 2^{n-m} decay of ϵ_n in standard QPE, with rectangular window and cutoff frequency at 2^{m-n-1} .

Table 2 Explicit forms of the DTFT of several windows in the cosine family

Window type	$\hat{w}(f)$	Decay like
Rectangular	$e^{-i\pi(M-1)f} \cdot \frac{1}{M} \cdot \frac{1}{\sin \pi f} \cdot \sin \pi M f$	$1/f$
Half sine	$-e^{-i\pi M f} \cdot \frac{\sin \frac{\pi}{2M}}{\sqrt{2M}} \cdot \frac{1}{\sin \pi f \sin \pi(f-\frac{1}{M})} \cdot \sin \pi M f$	$1/f^2$
Hann	$-e^{-i\pi M f} \cdot \sqrt{\frac{2}{3}} \frac{\sin^2 \frac{\pi}{2M}}{M} \cdot \frac{\cos \pi f}{\sin \pi(f+\frac{1}{M}) \sin \pi f \sin \pi(f-\frac{1}{M})} \cdot \sin \pi M f$	$1/f^3$

From the Table 2 we can predict that ϵ_n decays like $2^{3(n-m)}$ with the half sine window, and $2^{5(n-m)}$ with the Hann window. Deriving similar expressions for more complex window types of the family becomes tedious, so we turn to a result from Mallat [11], which says that if a bounded function $w(t)$ has bounded derivatives up to order p , then its energy spectrum decays faster than $1/f^{2(p+1)}$. We relate the order of derivatives to the number of 1's in $d_{R-}, \dots, d_{-1}, d_1, \dots, d_{R+}$, the windowing circuit parameters, and show that each additional 1 in the sequence adds to the order of derivatives by 1. Let $v(t) = w(t)(1 - e^{\pm i2\pi t/M})$, i.e. windows v and w differ by one "1" in their circuit parameter sequence. Assume w be infinitely differentiable in $(0, M)$, and r times differentiable at 0 and M . v is then infinitely differentiable in $(0, M)$. At endpoints 0 and M we compute

$$v'(t) = w'(t)(1 - e^{\pm i2\pi t/M}) + w(t)(1 - e^{\pm i2\pi t/M})' \quad (7)$$

The first term vanishes at the endpoints; the second term is at least r times differentiable, therefore v is $r + 1$ times differentiable at the endpoints, one more than w . $\square$

Back to our windows, a window function with r 1's in its windowing circuit parameters d will have asymptotic decay like $1/f^{r+1}$ in magnitude, or like $1/f^{2r+1}$ in total weight of spectral tail. The corresponding asymptotic marginal cost for 2x reliability (half fail rate) in qubit resource is $1/(2r + 1)$. This is 1/3, 1/5, 1/7, 1/9 for (half sine, Hamming), (Hann, Blackman), Hann^{1.5}, and Hann², respectively.

G. Estimating the efficiency of the windowing circuit

Windowing succeeds conditioned on measuring all $R = R_+ - R_-$ auxiliary qubits $\mathbf{a}$ to 1. Its probability equals the energy of the pre-measurement state in subspace $|\mathbf{a} = 2^R - 1\rangle$. In fact each of the 2^R subspaces contains a windowed version of $|x\rangle$ with its own window function, denoted w^a , $a = 0, \dots, 2^R - 1$, each of which takes its share of the total probability. A pre-measurement windowed substate $|w^a x\rangle$ has high energy if 1) w^a has high energy, and 2) magnitude sequences of w^a and x have high correlation. nding window energy, i.e. 37.5%.

Table 3 below shows the four window functions applied by the Hann windowing circuit in the pre-measurement subspaces. Windows in $|a_1 a_0 = 00\rangle$ and $|11\rangle$ have higher energy than the other two. Although the exact probabilities depend on $|x\rangle$, a good reference point is to assume $|x_j|$ distributes independent of j , which is satisfied by e.g. pure sinusoids (as is the case with QPE) and white noise. Under this assumption the probability of measuring $|11\rangle$ equals the corresponding window energy, i.e. 37.5%.

Table 3 Window functions in subspaces $|\mathbf{a} = \mathbf{a}_1 \mathbf{a}_0\rangle$ of the Hann windowing circuit

Subspace by $ a_1 a_0\rangle$	$ 00\rangle$	$ 01\rangle$	$ 10\rangle$	$ 11\rangle$
Function $w_j^p \propto$	$\frac{1}{2} + \frac{1}{2} \cos \frac{2\pi j}{M}$	$\frac{i}{2} \sin \frac{2\pi j}{M}$	$-\frac{i}{2} \sin \frac{2\pi j}{M}$	$\frac{1}{2} - \frac{1}{2} \cos \frac{2\pi j}{M}$
Graph				
Energy	0.375	0.125	0.125	0.375

For general window types of the cosine family the efficiency of the windowing circuit is quantized by the energy of the *unnormalized* window associated with $a = 2^R - 1$. Given parameters of the circuit $d_{R_-}, \dots, d_{-1}, d_1, \dots, d_{R_+}$, the unnormalized window can be computed in DFT domain by the following algorithm:

0. Initialize sequence $w \leftarrow (2^{-(R/2)})$.
1. For all $r = R_-, \dots, -1, 1, \dots, R_+$, do:
 - a. If $r < 0$, $w \leftarrow w * (-d_r - 1)/\sqrt{1 + d_r^2}$, where $*$ means convolution;
 - b. If $r > 0$, $w \leftarrow w * (1 - d_r)/\sqrt{1 + d_r^2}$.

The energy of w is then the energy of the window of interest by Parseval's theorem.

H. Short time Fourier transform circuit with overlapping windows

The following circuit implements STFT using Hann windows with window size 2^L and hop size 2^{L-M} .

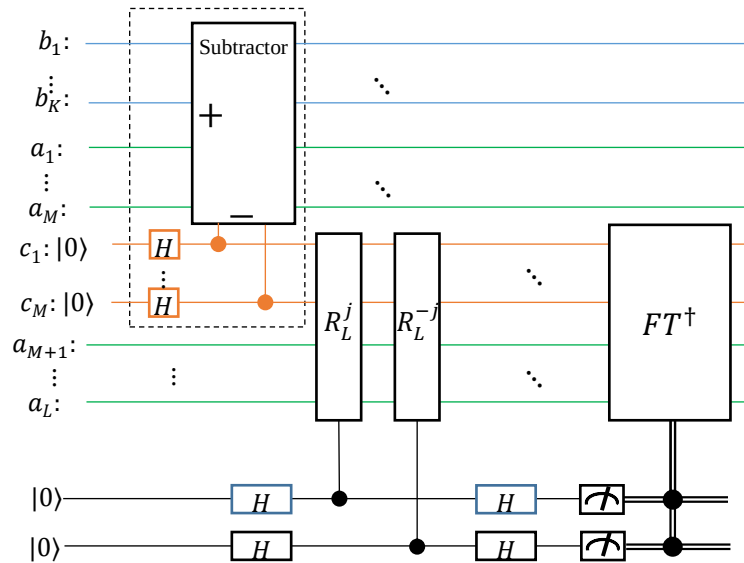


Figure 4 Circuit for STFT with overlapping windows

The input data is indexed by qubits $b_{1:K}, a_{1:L}$, with the interpretation of 2^K non-overlapping frames of length 2^L each. The bottom-right part of the circuit is easily identified as the Hann-windowed Fourier transform. The top-left subcircuit in the dashed frame is the framing circuit. The subtractor module implements invertible mapping $|b_{1:K}a_{1:M}, c_{1:M}\rangle \mapsto |(b_{1:K}a_{1:M} - c_{1:M}, c_{1:M})\rangle$, where the deduction by $c_{1:M}$ is modulo 2^{K+M} .

The circuit uses M auxiliary qubits to implement frame hop size of 2^{L-M} . There is no overlap between frames if $M = 0$, in which case the circuit simply computes 2^L -point DFT (notice $FT^\dagger$ would then be on $a_{1:L}$) on all 2^K blocks indexed by $b_{1:K}$, as expected. When $M > 0$, the auxiliary $c_{1:M}$ replaces $a_{1:M}$ as higher index bits within the frame; $a_{1:M}$ would then joint $b_{1:K}$ as the lower bits of frame index. The total number of frames is multiplied by 2^M , as expected.

Ignoring the auxiliary $c_{1:M}$ we may regard the higher register $b_{1:K}a_{1:M}$ as index to 2^{K+M} subframes of size 2^{L-M} , with subframe k in $|b_{1:K}a_{1:M} = k\rangle$. The group of Hadamard gates on $c_{1:M}$ takes them from 0 to equal superposition, effectually making 2^M copies of each subframe k , putting the j^{th} copy in subspace $|b_{1:K}a_{1:M} = k, c_{1:M} = j\rangle$. We now regard the higher register $b_{1:K}a_{1:M}$ as index to overlapping frames. The subtractor deducts $c_{1:M}$ from $b_{1:K}a_{1:M}$, which makes the j^{th} copy of the k^{th} subframe the j^{th} slot of the $(k - j)^{\text{th}}$ overlapping frame. Equivalently, that is making the k^{th} overlapping frame have the $(k + j)^{\text{th}}$ subframe in its j^{th} slot. Hence the k^{th} overlapping frame contains subframes from k to $k + 2^M$, or points from $k \cdot 2^{L-M}$ to $k \cdot 2^{L-M} + 2^L$, with $c_{1:M}$ as the higher index within the frame, as expected. $\square$